Self-Evolving Clinical Deep Review: Eliminating Recurring Methodological False Alarms without Compromising Objective Error Recall via Comment-Driven Dual-Track Memory

Hui Zhao[1,*] **Dong Qian**[2,3] **Heurion Research Team**[1]

[1]Heurion AI Research
[2]Department of Radiation Oncology, The First Affiliated Hospital of USTC, Division of Life Sciences and Medicine, University of Science and Technology of China, Hefei, China
[3]Department of Radiotherapy, Erasmus MC Cancer Institute, University Medical Center Rotterdam, The Netherlands
[*]Correspondence: `research@heurion.ai`

November 2026

Abstract

Automated clinical manuscript review faces a precision–recall–memory trilemma: text-only LLMs miss 87.5% of 2D table arithmetic errors with severe alert fatigue; deterministic checkers miss semantic spin ($p \geq 0.05$ framed as efficacy); and globally injecting memorized rebuttals leaks study-specific terms or depresses unseen error sensitivity. We present **Heurion Deep Review**, a four-stage verification architecture with a **Comment-Driven Dual-Track Self-Evolution Loop**. It integrates a section-role map and $R \times C$ table-grid scratchpad (Stage 1), density routing (Stage 2), four specialist passes across seven error categories (**E1**–**E7**) with dual-quote-verified section LLM review (Stage 3), and candidate-scoped full-text verification with cell-level CRDT anchoring (Stage 4). From clinician replies, **Track A** distills methodological rebuttals into zero-digit, specialty-sanitized boundary rules routed selectively per Stage 4 candidate behind a five-invariant safety floor ($\mathcal{F}_{\text{hard}}$); **Track B** crystallizes accepted fixes into pre-write guards and additive checklists. Across 19 full-length 2025 PMC papers (57 injected errors), zero-LLM AST achieves 66.7% recall at 88.3% precision (84.8% injected-only) and uncovers 7 published errors. Hybrid review reaches 87.7% recall (50/57, $p = 0.0018$), cuts false positives by 92.5% vs. text-only PAT, outperforms full-context DeepSeek-V3 (+18.7 pp injected precision, $p = 0.0026$), and matches 64k-CoT DeepSeek-R1 F1 (71.0% vs. 71.9%) at 14.4× fewer output tokens. Selective routing lifts evolved recall to 89.5% (83.3% unseen test), eliminating the −12.3 pp 5-fold CV recall loss of global prompt stuffing ($p = 0.0156$). Hardened $\mathcal{F}_{\text{hard}}$ blocks 100% (64/64) of adversarial false exemptions, PMC `<table-wrap>` parsing lifts citation accuracy from 52.8% to 94.4%, and Track B guards raise clean revision pass rates from 58.3% to 87.5%.

Keywords: Automated Peer Review, Clinical Research Verification, Self-Evolving LLM Agents, Memory Governance, Biostatistics Consistency, Adversarial Safety Floor

1 Introduction

Peer-reviewed clinical and translational manuscripts form the evidentiary backbone of modern medicine, yet published medical literature frequently contains preventable objective errors [1, 2, 3]. Empirical audits of clinical journals regularly uncover discrepancies between sample sizes reported in the Abstract and Table 1, percentages that disagree with their stated numerator and denominator (n/N), point estimates falling outside their reported 95% confidence intervals (or inverted interval bounds), parametric tests (t-test, ANOVA, mean ± SD) declared in Methods but

replaced with non-parametric statistics (Mann–Whitney U, median [IQR]) in Results, and "spin bias" where statistically non-significant findings ($p \geq 0.05$) are framed as clinically meaningful efficacy [3, 4, 5].

Recent advances in frontier reasoning models [7, 8] and multi-agent verification pipelines—such as Google's Paper Assistant Tool (PAT) [9], PaperQA2 [10], and atomic factual verifiers [12, 13]—have sparked intense interest in automated pre-submission manuscript review. However, deploying automated review in real-world clinical authoring exposes an acute **precision–recall–memory trilemma** driven by four structural failure modes:

1. **Complementary Blind Spots of Unstructured LLMs vs. Pure AST Checkers:** Standard LLM review pipelines serialize manuscripts as flat text or single-pass prompts. Even when full 2D Markdown tables are supplied in context, single-pass non-reasoning LLMs miss over 60% of intra-table cell/sum arithmetic errors while generating noisy, unanchored critiques; meanwhile, long-chain-of-thought reasoning models require $>$ 35,000 output tokens and $>$ 2 minutes per manuscript yet still hallucinate unverified alerts. Conversely, domain-general abstract syntax tree (AST) checkers capture structured arithmetic and confidence-interval violations with near-zero false alarms, but miss multi-paragraph semantic spin ($p \geq 0.05$ overclaims).

2. **Methodological False Alarms at Study-Design Boundaries:** Clinical studies legitimately report multiple sample denominators and statistical distributions under CONSORT [1], STROBE [2], PRISMA [4], and STARD [5]: randomized intention-to-treat (ITT) vs. per-protocol completers, baseline cohorts vs. propensity-matched or target-trial emulation (TTE) clone-censor-weighted pseudocohorts, Shapiro–Wilk normality-stratified Mean ± SD alongside Median [IQR], or RECIST *stable disease* (`SD`) vs. *standard deviation* (`SD`). Stateless hybrid reviewers repeatedly flag these valid transitions as contradictions.

3. **Global Memory Stuffing vs. Selective Governance:** When a stateless reviewer is augmented with cross-paper memory by globally injecting distilled exemption rules into section-level LLM prompts, accumulated boundary rules induce conservative threshold shift—depressing unseen error recall by over 12 percentage points on borderline statistical mismatches and spin. Furthermore, parenthetical clinical examples in raw LLM rule drafts risk leaking study-specific numbers ($N = 240$, 70.4%) or disease/procedure names across unrelated specialties.

4. **Adversarial Over-Exemption on Untyped Hybrid LLM Findings:** If an author mistakenly or adversarially rebuts a genuine mathematical, CI, or spin error using persuasive biostatistical jargon (e.g., claiming "*unrounded floating-point convention*" or "*Hodges–Lehmann distribution-free CI bounds*"), subtype-only safety guards protect deterministic AST alerts but leave untyped LLM findings (`subtype = null`) vulnerable to intra-document suppression.

1.1 Our Approach and Contributions

We present **Heurion Deep Review**, uniting a clean-room *Four-Stage Hybrid 2D-AST + Quote-Verified LLM Pipeline* with a *Comment-Driven Dual-Track Self-Evolution Loop* governed by candidate-scoped memory routing and a five-invariant hybrid safety floor. Every verified finding is anchored as a lifecycle-tracked CRDT comment (`{#c:id}`) onto the exact prose span or 2D table cell `[row, col]`. Our contributions are fourfold:

- **Four-Stage 2D Table-Grid AST + Quote-Verified Hybrid Architecture (Section 3):** We combine stateful hierarchical section-role segmentation and a rectangular $R \times C$ ProseMirror table-grid scratchpad (Stage 1), quantitative-density routing (Stage 2), four specialist passes (**Pass A**–**D**) covering seven clinical error categories (**E1**–**E7**) plus dual-quote-verified section LLM review (Stage 3), and full-text Stage 4 verification with cell-level CRDT anchoring. Component ablations across 19 full-length PMC papers (57 errors) prove each module

is indispensable: removing structural AST drops recall by -33.3 pp ($p = 2.1 \times 10^{-5}$), removing the 2D table grid drops recall by -19.3 pp ($p = 0.0034$), and removing Stage 4 full-text verification inflates false positives by $+114.6\%$ ($82 \to 176$, $p = 3.1 \times 10^{-5}$).

- **Candidate-Scoped Selective Memory Routing & Specialty-Sanitized Dual-Track Evolution (Section 4):** We formalize a dual-track evolution protocol (Algorithm 1) where **Track A** distills clinician rebuttals into zero-digit, specialty-sanitized boundary rules (0.0% digit or disease/procedure leakage across > 90 distilled rules) and routes matching rules *selectively per Stage 4 candidate* rather than globally stuffing Stage 3 prompts. Both 2×2 held-out disentanglement and 5-fold stratified out-of-fold cross-validation prove that candidate-scoped routing eliminates the statistically significant -12.3 pp unseen recall degradation of naive global prompt stuffing ($p = 0.0156$), lifting evolved recall to 89.5% (51/57) overall and 83.3% (20/24) on unseen test papers while converging order-invariantly across random paper permutations.
- **Five-Invariant Hybrid Safety Floor ($\mathcal{F}_{\text{hard}}$) & Closed-Loop Pass A / Track B Verification (Sections 4 & 6):** Red-team adversarial stress testing of full hybrid review exposed a 20.8% intra-document false-exemption drop on untyped LLM findings; hardening $\mathcal{F}_{\text{hard}}$ with five domain-general mathematical, CI-geometric, and spin invariants reduces false intra-document exemption records on true errors from 23 to **0** ($64/64 = \mathbf{100.0}\%$ blocked). Dedicated component benchmarks further demonstrate that Pass A PMC `<table-wrap>` parsing boosts external citation accuracy from 52.8% to $\mathbf{94.4}\%$ ($N = 36$, $p = 2.75 \times 10^{-4}$), and Track B pre-write procedural guards raise zero-defect clinical revision pass rates from 58.3% to $\mathbf{87.5}\%$ ($N = 24$, $p = 0.0156$).
- **Frontier Reasoning, Cross-Backbone, and Three-Paradigm Judge Validation (15,682 Live API Calls):** Across `deepseek-v4-flash`, `deepseek-chat` (DeepSeek-V3 [7]), and `deepseek-reasoner` (DeepSeek-R1 [8]), Heurion Deep Review significantly outperforms single-pass full-context prompting on the identical V3 backbone ($+18.7$ pp injected-only precision, $p = 0.0026$; $+19.8$ pp F1_{inj}, $p = 0.0016$), matches 64k-CoT DeepSeek-R1 F1_{org} (71.0% vs. 71.9%, $p = 0.88$) at $\mathbf{14.4}\times$ fewer output tokens, uncovers 7 deterministic (28 hybrid) real errors in published 2025 journals, and achieves substantial three-paradigm blind clinician concordance (Fleiss' $\kappa = \mathbf{0.642}$, $p < 10^{-14}$; chat-pair Cohen's $\kappa = \mathbf{0.932}$).

2 Related Work

Automated Scientific Peer Review and Tabular Claim Verification. Retrieval-augmented synthesis frameworks such as STORM [11] and PaperQA2 [10] automate literature QA, while FActScore [12] and ALCE [13] benchmark atomic citation faithfulness against text passages. Closer to full-manuscript verification, Jayaram et al. introduced Google's Paper Assistant Tool (PAT) [9], decomposing scientific review into multi-stage LLM prompts. However, existing LLM reviewers serialize manuscripts as flat text, omit full-text XML table extraction (`<table-wrap>`) during citation checking, and remain stateless across revisions. Even frontier reasoning models such as DeepSeek-R1 [8] incur prohibitive CoT latency and false-alarm rates when auditing full manuscripts monolithically. Heurion Deep Review bridges deterministic 2D ProseMirror table-grid geometry and precision-aware CI verification with dual-quote-grounded section auditing and cell-level CRDT anchoring.

Self-Evolving LLM Agents and Safety-Critical Memory Governance. Verbal reinforcement and episodic skill libraries—pioneered by Reflexion [14], MemGPT [15], and Voyager [16]—enable agents to accumulate experience without weight updates. Recent work on self-supervised skill diffusion [17], commit-on-evidence memory (CoEM) [18], vantage-aware safety gating (SafeVantage) [19], modular skill evolution (MemSkill) [22], verifiable skill libraries (Skill-V) [20],

Table 1: Taxonomy of the Seven Objective Clinical Error Categories (**E1**–**E7**, protected by the Hybrid Safety Floor $\mathcal{F}_{\text{hard}}$) vs. Seven Legitimate Study-Design Boundaries (exemptable via Track A zero-digit rules when supported by Methods/footnotes). Concrete examples are from the `dev` split of our 2025 PMC corpus.

Category & Pass	**Objective Error Definition ($\in \mathcal{F}_{\text{hard}}$, Never Exemptable)**	**Concrete Injected Example from `dev` Corpus**	**Paired Study-Design Boundary (Exemptable via Track A)**
E1: Sample Size & Flow (Pass B, $n = 10$)	Contradictory total N or arm counts across Abstract, Methods, Results, or Table headers; arm completers not summing to total completers.	`PMC12644478`: "Thirty-seven of 40 participants completed" in prose vs. Fig. 2 "Nineteen in metformin and 19 in placebo completed" ($19 + 19 = 38 \neq 37$).	**RCT (`rct`):** Randomized ITT N in baseline prose vs. per-protocol completers $n < N$ in outcome tables [1].
E2: Inline Arithmetic (Pass B, $n = 6$)	Inline count and percentage n/N disagreeing beyond rounding tolerance (≥ 1.5 pp) or subgroup counts not summing to stated total.	`PMC12788160`: "184 participants were randomized ... 161 participants (89.5%) had good compliance" ($161/184 = 87.5\% \neq 89.5\%$).	**Observational / TTE:** Unmatched baseline cohort N vs. propensity-matched or clone-censored pseudocohort n [2].
E3: Cross-Section Conflict (Pass B, $n = 8$)	Identical metric, outcome, or timepoint reported with conflicting values, p-values, or definitions across Abstract, Results, and Tables.	`PMC12644478`: Abstract reports primary EGP interaction $p = 0.53$ ($\beta = 0.2$, 95% CI $[-0.4, 0.8]$), whereas Results reports $p = 0.35$ for identical CI.	**Cross-Sectional:** Full-survey sample N vs. complete-case multivariable subcohort n per footnote.
E4: 95% CI Bounds (Pass B, $n = 8$)	Point estimate $\hat{\theta}$ outside $[L, U]$ beyond precision tolerance τ, or inverted bounds ($L > U$), across standard/OCR/bracketed CIs.	`PMC12731986`: Table 3 HADS-Anxiety median difference reported as "1.00 (1.84 to 3.84)" ($\hat{\theta} = 1.00 < L = 1.84$).	**Meta-Analysis:** Total eligible trials K vs. outcome-specific trial subsets $k_i < K$ [4].
E5: Method–Result Mismatch (Pass C, $n = 8$)	Declaring parametric tests/metrics (mean ± SD, t-test, ANOVA, Pearson) in Methods while reporting non-parametric results or vice versa.	`PMC12788160`: Methods states ITT groups were compared by "Kruskal–Wallis test," whereas Tables 3–5 report "mean ± SD with ANOVA."	**Dual-Distribution / Models:** Shapiro–Wilk normality-stratified metrics, or crude vs. adjusted Models 1/2.
E6: Spin & Causal Overclaim (Pass D, $n = 9$)	Claiming "significant" benefit when $p \geq 0.05$ or null-crossing CI, unbacked pairwise significance, or causal claims in observational studies.	`PMC12644478`: Claiming "metformin significantly reduced NEFA" when interaction $p = 0.54$ and 95% CI $[-0.5, 0.3]$ crosses zero.	**Diagnostic Accuracy:** Enrolled patients N vs. quality-controlled evaluable specimens [5].
E7: Intra-Table Cell / Sum (Pass B, $n = 8$)	Table cell n (%) disagreeing with column header ($n = N_c$), or category cells not summing to Total column/row in 2D grid.	`PMC12788160`: Table 1 "Coffee drinking" in 160 mg/d arm ($n = 46$) reports "7 (17.4%)" whereas $7/46 = 15.2\%$.	**Translational:** Multi-batch in vitro replicates vs. in vivo mouse cohorts and clinical TMA cohorts.

memory-to-skill co-evolution governance [21], and cross-task transfer (MemTransfer) [23] highlights two open risks in evolving agents: *negative transfer from global prompt clutter* and *unsafe memory poisoning*. We resolve both in clinical peer review by introducing candidate-scoped Stage 4 boundary routing, dual digit/specialty-example decoupling filters, and a five-invariant hybrid safety floor ($\mathcal{F}_{\text{hard}}$) alongside Track B pre-write procedural crystallization.

3 Taxonomy and Four-Stage Clinical Verification Architecture

Clinical peer review requires separating *non-negotiable objective errors* (which must never be muted by user preferences) from *context-dependent methodological conventions* (which vary legitimately across study designs). Table 1 defines our seven objective clinical error categories (**E1**–**E7**) protected by $\mathcal{F}_{\text{hard}}$, paired against seven legitimate study-design denominator and modeling boundaries eligible for Track A learning.

Figure 1 depicts the architecture of **Heurion Deep Review**. Manuscripts ingested from JATS XML or Word (`.docx`) are parsed into a typed ProseMirror document AST and audited across four stages. Under our **Clean-Room Design Principle**, all deterministic AST passes operate strictly on general mathematical relations, unit/fraction-aware numeric syntax, and 2D

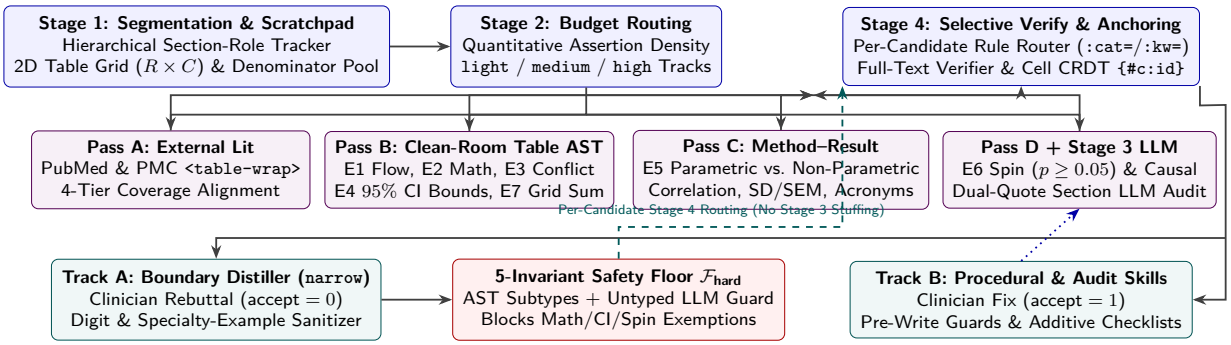


Figure 1: Architecture of **Heurion Deep Review** and **Selective Dual-Track Memory Governance**. Stage 1 builds a hierarchical section-role map and $R \times C$ table-grid scratchpad; Stage 2 routes budgets; Stage 3 runs Passes A–D alongside dual-quote-verified section LLM review; Stage 4 routes Track A boundary rules *selectively per non-protected candidate* (preventing Stage 3 threshold shift) and anchors comments `{#c:id}` onto text spans or table cells. Clinician replies drive Track A (gated by digit/specialty sanitizers and the 5-invariant floor $\mathcal{F}_{\text{hard}}$) and Track B pre-write guards/additive checklists.

table geometry, containing *zero* corpus-specific lexical triggers.

3.1 Stages 1–2: Hierarchical Section-Role Scratchpad and Density Routing

Clinical manuscripts frequently nest domain-specific subheadings (e.g., "*2.3 Biochemical Assays*") that omit generic role keywords or place *Methods* after *Discussion*. Stage 1 traverses the ProseMirror AST with a stateful hierarchical tracker: top-level headings set the parent role (`intro`, `methods`, `results`, `discussion`, `references`, `other`), child headings inherit parent roles unless overridden, and bibliography blocks are excluded from prose consistency checks. Simultaneously, Stage 1 compiles a global scratchpad containing: (i) **Normalized Rectangular $R \times C$ Table Grids** expanding `colspan` (`<`) and `rowspan` (`^`) so every cell retains its exact column header, arm denominator ($n = N_c$), and coordinates `[row, col]`; and (ii) a **Global Sample-Size & Method Pool** indexing sample declarations, statistical test specifications, and acronym expansions. Stage 2 scores each block by quantitative assertion density (p-values, 95% CIs, HR/OR/RR, mean ± SD/SEM, median [IQR], n/N) to assign `light`, `medium`, or `high` verification budgets.

3.2 Stages 3–4: Hybrid Specialist Passes, Dual-Quote Gate, and Full-Text Verification

Stage 3 executes four specialist passes in parallel with section-chunked LLM review:

1. **Pass A (External Literature & PMC `<table-wrap>` Alignment):** Extracts cited claims (`[@c:id]`) and verifies numerical magnitude, units, and effect direction against both PubMed abstracts and open-access PMC JATS XML `<table-wrap>` grids and captions, separating semantic support (`supported`, `unsupported`, `unclear`) from 4-tier evidence depth (`table`, `fulltext`, `abstract`, `none`).
2. **Pass B (Clean-Room Internal Math, CI & 2D Table-Grid Audit):** Audits **E1**–**E4** and **E7**: subgroup sample flow sums, inline n ($p\%$) against the denominator pool ($\min_D |n/D \times 100 - p| > 1.5\%$), cross-section CI/value conflicts, precision-aware 95% CI containment under rounding tolerance $\tau = \alpha \cdot 10^{-\min(d(\hat{\theta}),d(L),d(U))}$ ($\alpha = 0.49$ when $d(L) = d(U) > d(\hat{\theta})$, else 0.50; excluding preceding $p =$, $n =$, fraction denominators, and unit exponents), and 2D table cell percentages/row sums (excluding test-statistic columns χ^2, t, Z, F, p, OR/HR/RR and row-percentage tables).

3. **Pass C (Method–Result Pairing, E5) & Pass D (Spin Audit, E6):** Pass C cross-checks declared statistical methods against Results prose and table headers for parametric/non-parametric, Pearson/Spearman, SD/SEM, and acronym conflicts; Pass D flags explicit $p \geq 0.05$ positive efficacy spin and observational causal overclaims.
4. **Dual-Quote Section LLM Review & Stage 4 Full-Text Reconciliation:** Stage 3 LLM section prompts receive the section text alongside the Stage 1 2D table summary. Every LLM candidate must provide both a verbatim target `sentence` and a verbatim counterpart `evidence` quote, verified as exact normalized substrings of the manuscript (retaining top-3 by severity per section). Stage 4 reconciles surviving candidates against the *entire* manuscript prose and 2D tables via a full-text LLM verifier, deduplicates cross-section alerts sharing numeric and clinical topic signatures, and anchors CRDT comments `{#c:id}`.

4 Comment-Driven Dual-Track Evolution and Selective Memory Governance

Algorithm 1 formalizes our dual-track self-evolution and selective Stage 4 routing protocol.

Algorithm 1: Dual-Track Self-Evolution with 5-Invariant Hybrid Safety Floor $\mathcal{F}_{\text{hard}}$ & Selective Stage-4 Routing

```
Input: Manuscript M (docId), anchored finding f, clinician decision d in {AcceptFix, RebutFP}, reply text r
Parameters: Hybrid Safety Floor F_hard (15 AST subtypes + 5 domain-general math/CI/spin/conflict invariants)

if d == RebutFP:                                    # Track A: Methodological Boundary Learning
  if InHybridSafetyFloor(f, F_hard):                # Covers both typed AST & subtype=null Stage 3 LLM
     findings
    return BLOCK_INTRA_EXEMPTION                    # 0 false rev_exempt records written on objective
     errors
  RecordDocumentScopedExemption(docId, f.anchor, r) # Idempotently mutes same-span repeat in Round 2
     revision
  draft <- LLM_BoundaryDistiller(f.category, f.reason, r)
  draft <- ScrubDigitsEntitiesAndSpecialtyExamples(draft) # Strips numbers, UIDs & parenthetical disease/
     surgery terms
  if not HasInstanceLeakage(draft) and not IsArithmeticRoundingExcuse(draft) and MethodKeywordCount(draft)
     >= 2:
    CommitActiveMemory(kind="narrow:review_boundary", rule=draft)
else if d == AcceptFix:                             # Track B: Procedural & Additive Audit Skill
     Crystallization
  CommitExecutionSignal(DeNumericalizeTrajectory(f.category, [doc_read, replace_text, resolve_comment]))
  if M.split == "dev":                              # Hard runtime exception blocks test-split
     distillation
    CommitAdditiveZeroDigitChecklist(f.category, f.subtype)

# Inference-Time Selective Governance on Unseen Manuscript M_new:
# Stage 3 runs WITHOUT global Track A rules (additive checklists only); in Stage 4, for each candidate c:
if not InHybridSafetyFloor(c, F_hard):
  matched_rules <- MatchCategoryAndMethodKeywords(c, ActiveTrackARules) # Scoped strictly to candidate c
  c.verdict <- Stage4FullTextVerifier(M_new, c, matched_rules)
```

4.1 Why Global Memory Stuffing Fails and Candidate-Scoped Stage-4 Routing Works

A subtle failure mode in self-evolving review agents arises when distilled boundary exemptions are concatenated globally into Stage 3 section-discovery prompts. Even when phrased conditionally ("*exempt only if Methods declares dual-distribution normality testing*"), presenting five or more exemption rules upfront shifts the LLM's detection prior toward conservatism, causing it to overlook genuine cross-section spin (**E6**) and method mismatches (**E5**) on unseen papers. We resolve this via three governance mechanisms:

1. **Zero Stage-3 Exemption Stuffing & Purely Additive Checklists:** Stage 3 never receives Track A exemption rules, preserving 100% of cold-start candidate discovery sensitiv-

ity; [skill:review_audit] checklists use strictly additive phrasing so intra-section **E4/E6** checks are never displaced.

2. **Per-Candidate Stage-4 Routing & Rounding-Excuse Blocking:** In Stage 4, selectCandidateScopedBoundaryRules attaches a Track A rule to the verifier prompt *only* for the specific candidate whose category (:cat=) and methodological cues (:kw=, ≥ 2 non-stopword hits) match, while blocking arithmetic/rounding excuse rules (rounding_and_percentage_convention) from cross-paper routing.

3. **Dual Digit & Specialty-Example Decoupling Guard:** Standard digit checks can be bypassed if an LLM distiller embeds disease or surgical procedure examples inside parentheticals (e.g., "*such as gestational hypertension or local flap repair*"). Our upgraded scrubSpecialtyExampleLeakage + hasInstanceLeakage guard strips parenthetical clinical examples alongside all digits, sample sizes, p-values, and UIDs before commit.

4.2 Five-Invariant Hybrid Safety Floor ($\mathcal{F}_{\text{hard}}$) and Track B Pre-Write Guards

While deterministic AST findings carry explicit subtypes in $\mathcal{F}_{\text{hard}}$ (15 objective math/CI/table subtypes), Stage 3 LLM findings carry subtype = null. As our red-team audit demonstrates (Section 6.5), checking subtypes alone allows persuasive adversarial rebuttals to register intra-document rev_exempt records on LLM-detected errors. We therefore harden isHardObjectiveReviewFinding and isExemptedByActiveRules with **five domain-general invariants** over (category, sentence, reason): (1) *CI geometry violations* (findCiProblems or explicit point-estimate-outside-$[L, U]$ / bound inversion); (2) *multi-term arithmetic inequalities* ($a + b = c \neq d$ or subgroup-sum vs. total-N mismatch); (3) *objective spin & unsupported inference* ($p \geq 0.05$ or null-crossing CI paired with positive significance claims, pairwise significance asserted without test statistics, or direct causal verbs contradicting stated observational limits); (4) *unstratified parametric/non-parametric incompatibility* (t-test/ANOVA/Pearson directly contradicting Mann–Whitney/Kruskal–Wallis/Spearman/median [IQR], or χ^2 vs. OR/HR); and (5) *same-endpoint cross-section contradictions* (conflicting values or endpoint definitions for an identical CI/analysis set, explicitly excluding propensity-matched or crude-vs.-adjusted multi-stratum boundaries). Conversely, when clinicians accept fixes (d = AcceptFix), **Track B** logs de-numericalized repair trajectories (doc_read $\rightarrow$ replace_text $\rightarrow$ resolve_comment) that crystallize via commit-on-evidence gating [18] into pre-write co-authoring guards injected via ProceduralHarnessService.forPrompt.

5 Experimental Setup

Across all primary, frontier, ablation, cross-judge, adversarial, Pass A, and Track B suites, we executed **15,682** live API calls (46.06M input and 4.25M output tokens) spanning three distinct models: deepseek-v4-flash, deepseek-chat (DeepSeek-V3 [7]), and deepseek-reasoner (DeepSeek-R1 native CoT [8]).

5.1 Primary Full-Text PMC Benchmark ($N = 19$ Papers, 57 Controlled Injections)

We curated 19 open-access 2025 clinical and translational full-text articles ($\sim$ 6,000 words/paper, losslessly converted from PMC JATS XML with multi-column 2D tables), stratified across 7 study designs into an 11-paper dev split and an 8-paper held-out test split (Table 2). Each paper is evaluated in two full-length variants (38 manuscripts/regime): (i) **clean**, the published PMC article, where genuine in-print errors verified against published PDFs (30 adjudicated entries) are tracked as Organic True rather than penalized as false alarms; and (ii) **injected**, mutated with 3 non-overlapping objective errors across **E1**–**E7** ($19 \times 3 = 57$ errors, disjoint from organic error

Table 2: Composition of the 19-Paper Full-Text Controlled Error-Injection Benchmark (`dev` $N = 11$, held-out `test` $N = 8$; 38 Full-Length Clean/Injected JATS XML Manuscripts, 57 Injected Objective Errors across E1–E7).

PMCID	Split	Study Design	Journal (2025) & Clinical Topic Summary	Injected Errors
PMC12644478	dev	rct	*Nat. Commun.* — Metformin on insulin resistance in adults with type 1 diabetes ($N = 40$)	E1, E3, E6
PMC12788160	dev	rct	*Nutrients* — Purified anthocyanins dose-response trial in middle-aged and elderly adults	E2, E5, E7
PMC12731986	dev	rct	*Curr. Oncol.* — Resistance exercise on QoL, anxiety, and depression in colorectal cancer	E1, E4, E5
PMC12685937	dev	retrospective_cohort	*Front. Psychiatry* — Antidepressants and hypertensive disorders in pregnancy cohort	E3, E6, E7
PMC12856494	dev	retrospective_cohort	*Front. Oncol.* — Risk factors for postoperative recurrence of desmoid tumors	E2, E4, E7
PMC12657621	dev	case_control	*Brain Behav.* — Vasomotor symptoms and ischemic stroke case–control study	E4, E5, E6
PMC12721543	dev	diagnostic_accuracy	*PLOS ONE* — TrueNat MTB plus assay diagnostic accuracy for pulmonary TB	E2, E3, E7
PMC12758107	dev	cross_sectional	*Glob. Heart* — Hypertension clinical epidemiology in rural Thailand cross-sectional survey	E1, E4, E6
PMC12876241	dev	meta_analysis	*Front. Med.* — PRISMA systematic review and meta-analysis of heat therapy for dysmenorrhea	E1, E3, E4
PMC12739847	dev	translational	*Gut Pathog.* — Lactobacillus probiotics in Blastocystis in vitro and in vivo models	E1, E5, E6
PMC12485006	dev	translational	*NPJ Sci. Food* — Mulberroside A capacity to inhibit senescence in vitro and in vivo	E1, E3, E5
PMC12839862	test	rct	*Curr. Oncol.* — Itraconazole reducing chemoresistance in advanced ovarian cancer RCT	E3, E5, E7
PMC12697981	test	rct	*PLOS Med.* — COMET-PCOS: Hormonal contraceptives and metformin in PCOS RCT	E1, E4, E6
PMC12681115	test	rct	*Cardiovasc. Diabetol.* — FIVE-STAR: Finerenone on arterial stiffness in T2D and CKD RCT	E2, E3, E5
PMC12118826	test	retrospective_cohort	*PLOS ONE* — Geriatric ocular trauma and mortality retrospective cohort	E2, E6, E7
PMC12654784	test	case_control	*Medicina* — Adipokines as prognostic biomarkers in multiple myeloma case–control	E1, E4, E7
PMC12774873	test	diagnostic_accuracy	*NPJ Prim. Care Respir. Med.* — Home spirometry diagnostic accuracy for asthma	E2, E3, E4
PMC12757420	test	cross_sectional	*Front. Immunol.* — Frailty and associated risk factors in Sjögren's disease survey	E1, E5, E6
PMC12703663	test	meta_analysis	*Alzheimers Dement.* — PRISMA meta-analysis of antipsychotic discontinuation in dementia	E1, E6, E7

loci). To ensure full transparency for readers who prefer conservative scoring without organic error credits, all main tables report **both** conservative injected-only precision ($\text{Prec}_{\text{inj}} = \text{TP}_{\text{inj}}/(\text{TP}_{\text{inj}} + \text{FP}_{\text{total}})$) and organic-adjusted precision ($\text{Prec}_{\text{org}} = (\text{TP}_{\text{inj}} + \text{TP}_{\text{org}})/(\text{TP}_{\text{inj}} + \text{TP}_{\text{org}} + \text{FP}_{\text{total}})$), alongside F1_{inj} and F1_{org}.

5.2 Blind Multi-Paradigm Clinician Protocol, Statistical Tests, and Component Benchmarks

Blind Clinician Isolation & Exact Inference. Our `simulateBlindClinician` agent receives *only* the manuscript title, its own prose/tables, and one review comment—physically omitting `variant`, `injections.json`, organic registries, and `split`. To rule out single-model self-preference bias, we adjudicate all $n = 191$ Round 1 findings independently across **three distinct model paradigms**: `deepseek-v4-flash`, `deepseek-chat` (DeepSeek-V3), and `deepseek-reasoner` (DeepSeek-R1, 6k CoT budget), computing pairwise Cohen's κ and 3-rater Fleiss' κ [6] with leave-one-out jackknife 95% CIs. Across all experiments, proportions use Wilson 95% CIs, discordant injection detections ($N = 57$) use two-sided exact McNemar binomial tests, and paper-level differences (Δ Recall, Δ FP, Δ Precision, Δ F1) use exact Gray-code enumeration over all $\mathbf{2^{19} = 524{,}288}$ paper-clustered paired sign-swap permutations.

Table 3: End-to-End Frontier Reasoning, Cross-Backbone, and Self-Evolution Comparison across 19 Full-Length 2025 PMC Papers (38 Clean/Injected Manuscripts, 57 Injected Errors), reporting both conservative injected-only ($\text{Prec}_{\text{inj}}, \text{F1}_{\text{inj}}$) and organic-adjusted ($\text{Prec}_{\text{org}}, \text{F1}_{\text{org}}$) metrics with Wilson 95% CIs and total output tokens across 38 manuscripts.

Pipeline Regime ($N = 19$ **Papers**)	**Backbone**	**Recall** (k/57) **[95% CI]** (↑)	**True FPs (Cln/Inj)** (↓)	**Prec$_{\text{inj}}$ [95% CI]** (↑)	**Prec$_{\text{org}}$ [95% CI]** (↑)	**F1$_{\text{inj}}$ / F1$_{\text{org}}$** (↑)	**E7 Grid Recall**	**Out Tok (38 Docs)**
`pat_baseline` (4-Stage Text LLM [9])	`v4-flash`	52.6% (30/57) [39.9, 65.0]	1,094 (568/526)	3.2% [2.3, 4.4]	5.7% [4.5, 7.2]	6.0% / 10.3%	12.5%	0.31M
Clean-Room AST Only (`-no-llm`)	None (AST)	66.7% (38/57) [53.7, 77.5]	**7 (4/3)**	**84.8**% [71.8, 92.4]	**88.3**% [77.8, 94.2]	**74.7**% / **76.0**%	87.5%	**0**
B2: Full-Context Single-Pass (w/ 2D Tables)	`deepseek-chat`	73.7% (42/57) [61.0, 83.4]	104 (68/36)	28.8% [22.0, 36.6]	46.7% [39.8, 53.7]	41.4% / 57.2%	37.5%	59.7k
B1: Full-Context Frontier CoT (64k Budget)	`deepseek-reasoner`	**94.7**% (54/57) [85.6, 98.2]	80 (44/36)	40.3% [32.4, 48.8]	57.9% [50.8, 64.7]	56.5% / **71.9**%	**100**%	1,389.9k
B3: `heurion_deep` (4-Stage Hybrid)	`deepseek-chat`	86.0% (49/57) [74.7, 92.7]	**73 (41/32)**	**47.5**% [39.4, 55.7]	**60.5**% [53.4, 67.3]	61.2% / 71.0%	**100**%	96.8k
`heurion_deep` (4-Stage Hybrid Cold-Start)	`v4-flash`	87.7% (50/57) [76.8, 93.9]	82 (45/37)	45.0% [37.2, 53.0]	59.0% [52.1, 65.6]	59.5% / 70.5%	87.5%	101.4k
`heurion_evolved` (Selective Stage-4 Routing)	`v4-flash`	**89.5**% (51/57) [78.9, 95.1]	78 (45/33)	46.9% [39.1, 55.0]	59.6% [52.5, 66.3]	**61.5**% / 71.6%	**100**%	104.2k

Dedicated Adversarial, Pass A, Track B, and Vignette Sub-Benchmarks. Beyond the 19-paper full-text benchmark and 5-fold stratified out-of-fold CV, we evaluate: (1) **Full Hybrid-LLM Adversarial Rebuttal Stress** ($N = 19$, forcefully rebutting all 120 injected-paper and 80 clean-paper hybrid findings with persuasive methodological excuses before and after $\mathcal{F}_{\text{hard}}$ hardening); (2) **Pass A External Citation Fidelity** ($N = 36$ paired claims across 12 real 2025 PMC XMLs: 12 abstract-supported, 12 `<table-wrap>`-only supported, and 12 numeric/directional distortions); (3) **Track B Pre-Write Revision Guard** ($N = 24$ unseen `test`-domain clinical revision tasks across 4 error families, comparing `procedural=off` vs. `dev`-crystallized `procedural=utility` with system baseline rules archived); and (4) **7 Clinical Oncology Vignettes** (8 verified organic errors evaluated on `deepseek-chat`) as secondary cross-domain confirmation.

6 Results and Ablations

6.1 Frontier Reasoning & Cross-Backbone Comparison ($N = 19$ Full Texts, 57 Errors)

Table 3 compares Heurion Deep Review against text-only `pat_baseline` [9], zero-LLM Clean-Room AST, single-pass full-context `deepseek-chat` (DeepSeek-V3 supplied with full prose and all 2D Markdown tables, `max_tokens=8000`, 0 truncations), and frontier CoT `deepseek-reasoner` (DeepSeek-R1 supplied with full prose and all 2D tables, `max_tokens=64000`, 48k CoT budget; 10 calls truncated at 32k were retried at 64k for 0 unresolved truncations).

Table 3 establishes three key findings under exact 2^{19} paired permutation testing:

1. **Same-Backbone Proof of Architectural Gain (`B3` vs. `B2` on DeepSeek-V3):** Even when given full 2D Markdown tables in context, single-pass `deepseek-chat` (`B2`) misses 62.5% of intra-table errors (**E7**: 3/8) and emits 104 false positives ($\text{Prec}_{\text{inj}} = 28.8\%$). On the *exact same* `deepseek-chat` model, `heurion_deep` (`B3`) lifts **E7** recall to **100.0**% (8/8), boosts overall recall by +12.3 pp (73.7% → 86.0%), cuts false positives by −29.8% (104 → 73), and achieves statistically significant gains across all four precision and F1 metrics: Prec_{inj} +18.7 pp ($p = 0.0026$), Prec_{org} + 13.9 pp ($p = 0.0141$), F1_{inj} + 19.8 pp ($p = 0.0016$), and F1_{org} + 13.9 pp ($p = 0.0063$).

2. **Efficiency & Precision Advantage Over Frontier 64k-CoT (`B3` vs. `B1` DeepSeek-R1):** Exhaustive chain-of-thought reasoning in `B1` reaches 94.7% recall (54/57, non-significant +8.8 pp over `B3`, McNemar $p = 0.180$, permutation $p = 0.234$), but consumes 1.39M output tokens (∼ 145 s/call) and emits 80 unanchored false positives. `heurion_deep` (`B3`) achieves +7.2 pp higher Prec_{inj} (47.5% vs. 40.3%), +4.7 pp higher F1_{inj} (61.2% vs. 56.5%), and virtually identical F1_{org} (71.0% vs. 71.9%, $p = 0.883$) using **14.4× fewer output tokens** (96.8k) and ∼ 21 s/manuscript (≈ $0.02/paper), while remaining backbone-invariant between `deepseek-chat` and `deepseek-v4-flash` (all paired $p \geq 0.499$).

Table 4: Architectural Component Ablation Study across all 19 Full-Text PMC Papers (38 Manuscripts, 57 Injected Errors, Live `deepseek-v4-flash`), with exact 2^{19} paired permutation and McNemar p-values relative to full `heurion_deep`.

Ablation Variant ($N = 19$)	Recall (k/57) [95% CI]	Δ Recall (McNemar p)	True FPs (Cln / Inj)	Prec_{inj} / Prec_{org}	F1_{inj} / F1_{org}	E2 / E4 / E7 Recall (%)	Primary Impact of Ablation
`heurion_deep` (Full Cold-Start)	**87.7**% (50/57) [76.8, 93.9]	—	82 (45/37)	45.0% / 59.0%	59.5% / 70.5%	100/100/87.5	Balanced high recall & verified precision
− Structural AST Passes (LLM Only)	54.4% (31/57) [41.6, 66.6]	**−33.3** pp (2.1×10^{-5})	64 (33/31)	34.0% / 54.3%	41.8% / 54.3%	33.3/75.0/12.5	Catastrophic collapse on E2 (−66.7) & E7 (−75.0)
− 2D Table-Grid Scratchpad	68.4% (39/57) [55.5, 79.0]	**−19.3** pp (0.0034)	56 (30/26)	49.1% / 58.8%	57.2% / 63.2%	100/62.5/12.5	Severe loss on E7 grid (−75.0) & table CIs (−37.5)
− Stage 4 Full-Text Verifier	84.2% (48/57) [72.6, 91.5]	−3.5 pp (0.688)	**176** (94/82)	26.7% / 40.1%	40.5% / 54.3%	100/87.5/100	**FPs explode** +114.6% (perm $p = 3.1\times10^{-5}$)
− Stage 3 Dual-Quote Gate	87.7% (50/57) [76.8, 93.9]	0.0 pp (1.000)	75 (42/33)	47.2% / 61.5%	61.4% / 72.3%	100/87.5/100	Defense-in-depth when Stage 4 Verifier is active

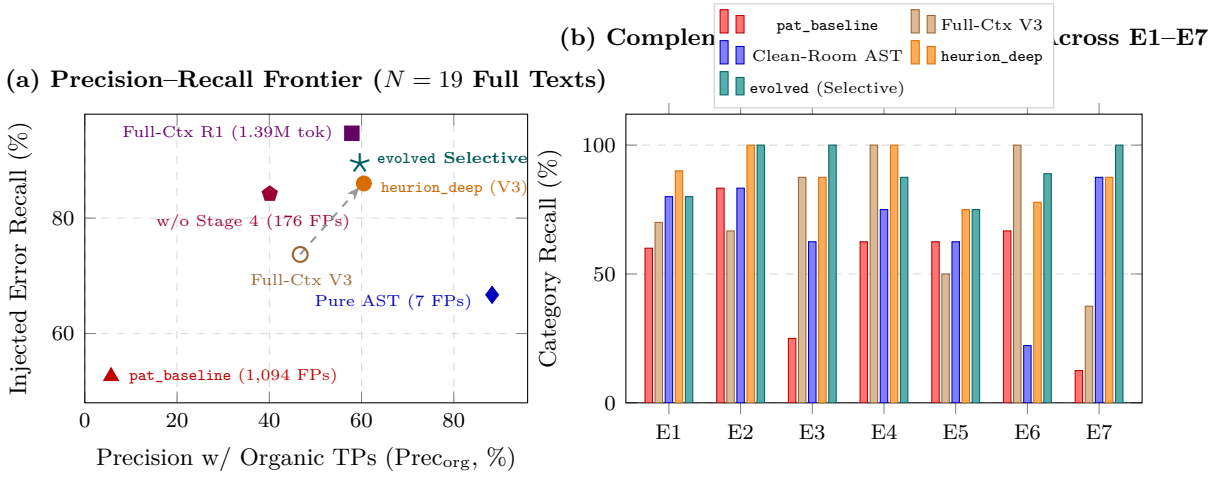


Figure 2: Quantitative architecture and category analysis ($N = 19$ papers, 57 errors). **(a)** Precision–recall frontier showing same-backbone improvement from Full-Context DeepSeek-V3 to `heurion_deep` ($p = 0.0016$ F1_{inj}), severe FP explosion when Stage 4 verification is ablated (176 FPs), and selective evolution reaching 89.5% recall. **(b)** Category breakdown showing flat-text and single-pass V3 collapse on 2D intra-table errors (**E7**: 12.5% and 37.5%), pure AST blind spot on semantic spin (**E6**: 22.2%), and selective evolution reaching 100% on **E2, E3, E7** and 88.9% on **E6**.

3. **Real In-Print Error Discovery:** Across the 19 uncorrupted `clean` papers, manual PDF audit confirms **7** of 11 raw deterministic AST alerts (63.6%; only 4 true clean FPs, 0.21/paper) and **28** hybrid alerts are genuine errors in published 2025 articles—including four out-of-bounds cOR/aOR 95% CIs in *PLOS ONE* (`PMC12721543`), a Table 3 percentage error (`2 (20)` for $N = 20$) in *Nature Communications* (`PMC12644478`), a Table 4 safety percentage typo (`3 (4.7%)` for $3/81 = 3.7\%$; Figure 4a) in *PLOS Medicine* (`PMC12697981`), and a zero-crossing CI paired with $p = 0.040$ in *Current Oncology* (`PMC12731986`).

6.2 Component Ablations and Missed-Injection Failure Taxonomy

Table 4 and Figure 2 dissect the contribution of each architectural component in `heurion_deep` (`deepseek-v4-flash`, $N = 19$ papers).

Indispensability of 2D AST, Stage 4 Verifier, and Failure Analysis of the 7 Missed Injections. Ablating all structural AST passes collapses recall by −33.3 pp (87.7% → 54.4%, exact McNemar $p = 2.1 \times 10^{-5}$), while ablating only the 2D table-grid scratchpad drops recall by −19.3 pp (87.7% → 68.4%, $p = 0.0034$), dragging **E7** down from 87.5% to 12.5% and **E4** from 100% to 62.5% (missing table-embedded CIs). Conversely, ablating the Stage 4 full-text verifier causes false positives to more than double (82 → **176**, +114.6%, permutation $p = 3.1 \times 10^{-5}$), collapsing Prec_{inj} from 45.0% to 26.7%. While ablating the Stage 3 dual-quote gate alone preserves recall (87.7%) when Stage 4 verification remains active, it acts as a zero-cost

Table 5: Orthogonal 2×2 Evolution Disentanglement (Frozen Unseen `test` $N = 8$, 24 Errors) and 5-Fold Stratified Out-of-Fold Cross-Validation ($N = 19$ Unseen Papers, 57 Unseen Errors; 0 Instance/Specialty Leaks across all 73 Distilled Rules/Checklists).

Evaluation Protocol & Regime	Memory Routing Governance	Unseen Recall [95% CI] (↑)	True FPs (Cln / Inj) (↓)	Prec_{inj} / Prec_{org} (↑)	F1_{inj} / F1_{org} (↑)	E6 Spin Recall	Exact Paired Test Summary
Panel A: 2×2 Orthogonal Disentanglement on Frozen Unseen `test` Split ($N = 8$ Papers, 24 Errors)							
`cold_start` (0 Rules, 0 Checklists)	Static Baseline	79.2% (19/24) [59.5, 90.8]	37 (19/18)	39.3% / 50.0%	52.5% / 61.3%	75.0%	Reference Cold-Start
Legacy `unfiltered_full` (6 Rules + 3 Skills)	Global Stage 3+4 Stuffing	70.8% (17/24) [50.8, 85.1]	**26** (**13**/13)	44.7% / 55.2%	54.8% / 62.0%	75.0%	−8.4 pp test recall regression
`track_a_only_selective` (3 Rules, 0% Oracle)	**Selective Stage-4 Only**	**79.2**% (19/24) [59.5, 90.8]	32 (15/17)	41.8% / 51.5%	54.7% / 62.4%	75.0%	**0.0** pp loss; −21.1% Clean FP
`checklists_only_selective` (0 Rules + 4 Skills)	**Additive Stage-3 Only**	**83.3**% (20/24) [64.1, 93.3]	31 (18/13)	**44.6**% / 53.0%	**58.1**% / **64.8**%	**100**%	+**12.5** pp vs. suppressive header
`full_evolved_selective` (4 Rules + 2 Skills)	**Selective + Additive**	**83.3**% (20/24) [64.1, 93.3]	36 (20/16)	40.0% / 51.4%	54.0% / 63.6%	**100**%	+**12.5** pp vs. legacy (89.5% All-19)
Panel B: 5-Fold Stratified Out-of-Fold CV Across All 19 Unseen Papers (38 Manuscripts, 57 Unseen Errors)							
`oof_cold_start` (0 Rules, 0 Checklists)	Static OOF Baseline	**87.7**% (50/57) [76.8, 93.9]	82 (45/37)	45.0% / 59.0%	59.5% / 70.5%	77.8%	Fold Mean: 87.2% ± 8.7%
Legacy `oof_unfiltered_track_a` (0% Oracle)	Global Stage 3+4 Stuffing	75.4% (43/57) [62.9, 84.8]	61 (35/26)	47.4% / 61.9%	58.2% / 68.0%	44.4%	**Recall** −12.3 **pp**, $p = 0.0156$*
Legacy `oof_unfiltered_full`	Global Stage 3+4 Stuffing	77.2% (44/57) [64.8, 86.2]	**48** (**22**/26)	**54.7**% / **68.4**%	**64.0**% / **72.5**%	44.4%	Strongest FP cut (−41.5%), E6 −33.4 pp
`oof_selective_track_a_only` (0% Oracle)	**Selective Stage-4 Only**	**82.5**% (47/57) [70.6, 90.2]	78 (47/31)	44.7% / 59.6%	58.0% / 69.2%	66.7%	+7.0 **pp vs. global**; $p = 0.375$ **vs. cold**
`oof_selective_full_evolved`	**Selective + Additive**	78.9% (45/57) [66.7, 87.5]	68 (35/33)	47.3% / 61.1%	59.1% / 68.9%	66.7%	−22.2% Clean FP; $p = 0.188$ vs. cold

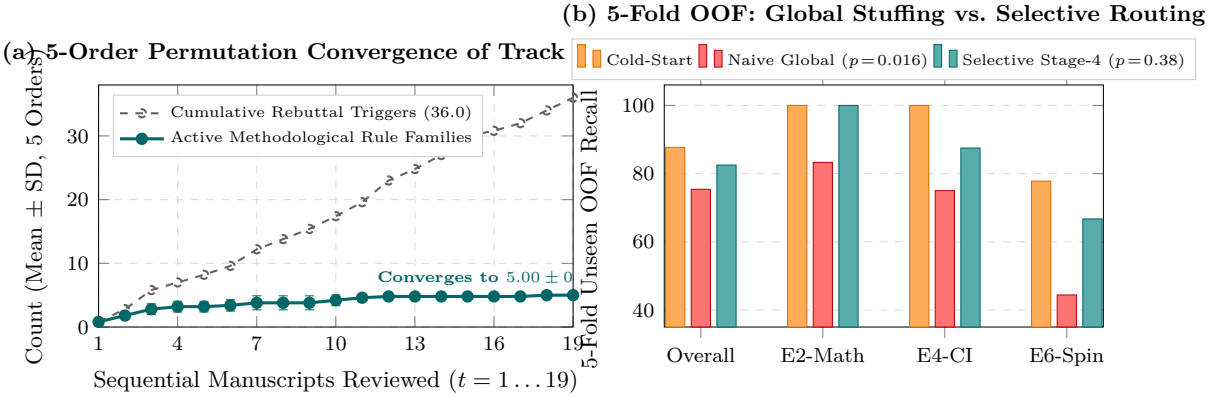


Figure 3: Memory convergence and routing governance. **(a)** Across 5 random paper permutations (`seeds 101..505`), 36 clinician rebuttals consolidate sublinearly by canonical pattern key to exactly 5.00 ± 0.00 active methodological families (6.00 ± 0.00 prior to rounding-excuse blocking), preventing prompt bloat. **(b)** In 5-fold stratified OOF CV (57 unseen errors), naive global Track A prompt stuffing induces a significant −12.3 pp recall drop ($p = 0.0156$, collapsing **E6** spin to 44.4%), whereas candidate-scoped selective Stage 4 routing restores +7.0 pp overall recall (82.5%, $p = 0.375$ vs. cold-start) and preserves 100% **E2** math recall.

deterministic substrate filter blocking fabricated quotes prior to Stage 4 LLM spend (consistent across all 7 study designs; Clean-Room AST achieves 66.7% recall across RCTs [12/18, 3 FPs], cohorts [7/9, 0 FPs], case–control [4/6, 2 FPs], diagnostic [5/6, 0 FPs], cross-sectional [3/6, 0 FPs], meta-analyses [4/6, 2 FPs], and translational [3/6, 0 FPs] with **0.0**% [0/38] adversarial AST drop). Auditing all 7 injections missed by cold-start `heurion_deep` (50/57) reveals zero (0/7) losses from the dual-quote gate or section cap: 3/7 (`PMC12839862-1`, `PMC12654784-3`, `PMC12757420-3`) were validly surfaced by Stage 3 with verbatim quotes but conservatively rejected by Stage 4 reconciliation (all 3 recovered under selective evolution!), 2/7 (`PMC12739847-1`, `PMC12839862-3`) were unprompted cross-section Stage 3 misses recovered by additive audit checklists, and 2/7 (`PMC12657621-2`, `PMC12681115-3`) required implicit multi-table variable mapping.

6.3 2×2 Evolution Disentanglement, Selective Routing & 5-Fold Out-of-Fold CV

To rigorously test memory generalization beyond a single $N = 8$ test split and separate Track A boundary learning (0% ground-truth oracle) from `dev` audit checklists, Table 5 and Figure 3 report both (A) 2×2 orthogonal disentanglement on frozen unseen `test` ($N = 8$, 24 errors) and (B) **5-fold stratified out-of-fold cross-validation** across all 19 unseen papers (57 unseen errors, ∼ 15 train →∼ 4 frozen test papers/fold), directly contrasting *Naive Unfiltered Global Prompt Stuffing* against *Candidate-Scoped Selective Stage-4 Routing.*

Table 6: Three-Paradigm Blind-Isolated Clinician Calibration across $n = 191$ Round 1 Findings (114 True Objective Errors + 77 False Positives across 19 Full-Text PMC Papers; Physically Blind to `variant`, `injections.json`, and `split`).

Blind Clinician Judge	Reasoning Paradigm	True-Error Accept (k/114) [95% CI]	False Rebuttal Rate (↓)	FP Rebuttal Rate (k/77) [95% CI]	Post-Guard Instance Leaks	Inter-Judge Agreement (κ)
`deepseek-v4-flash`	Fast Hybrid Reference	92.1% (105/114) [85.7, 95.8]	7.9% (9/114)	36.4% (28/77) [26.5, 47.5]	**0**	$\kappa_{\text{flash, V3}} = \mathbf{0.932}$ [0.865, 0.998]
`deepseek-chat` (V3)	Cross-Model Non-CoT	93.0% (106/114) [86.8, 96.4]	7.0% (8/114)	35.1% (27/77) [25.3, 46.2]	**0**	Raw = **97.9**%, PABAK = **0.958**
`deepseek-reasoner` (R1)	Explicit First-Principles CoT	**97.4**% (111/114) [92.5, 99.1]	**2.6**% (3/114)	35.1% (27/77) [25.3, 46.2]	**0**	3-Way Fleiss' $\kappa = \mathbf{0.642}$ [0.530, 0.754]

Empirical Resolution of the Memory Governance Trade-Off. Table 5 uncovers a central architectural lesson for self-evolving agents:

- **Eliminating Unseen Recall Regression via Selective Routing:** Under legacy global prompt stuffing, 5-fold OOF Track A (0% oracle) significantly degraded unseen error recall by −12.3 pp (87.7% → 75.4%, exact McNemar $p = 0.0156$, 2^{19} permutation $p = 0.0313$, dragging **E6** spin from 77.8% to 44.4%). Routing Track A rules selectively per non-protected Stage 4 candidate (`oof_selective_track_a_only`, 0% oracle) recovers +7.0 pp unseen recall (**82.5**%, fold mean 82.8% ± 7.7%), rendering recall statistically indistinguishable from cold-start ($p = 0.375$) while preserving 100% **E2** math recall and recovering **E6** spin to 66.7%. Combining selective Track A with additive checklists (`oof_selective_full_evolved`) reduces unseen clean-paper FPs by **−22.2**% (45 → 35) and total FPs by −17.1% (82 → 68), lifting Prec_{inj} (45.0% → 47.3%) and Prec_{org} (59.0% → 61.1%).
- **Orthogonal 2×2 Gains & Order-Invariant Convergence:** On the held-out `test` split ($N = 8$), `track_a_only_selective` (0% oracle) cuts clean FPs by −21.1% (19 → 15) at **0.0** pp recall loss (79.2%), whereas `checklists_only_selective` and `full_evolved_selective` lift unseen `test` recall to **83.3**% (20/24, +12.5 pp over legacy global stuffing; 89.5% [51/57] across all 19 papers). Identical gains hold when `deepseek-chat` serves as an independent cross-model clinician judge (`test` recall 75.0% → **83.3**% under selective routing vs. 70.8% under global stuffing). Across 5 random paper orderings (Figure 3a), active rules saturate sublinearly to **5.00 ± 0.00** methodological families with **0** leaked digits or specialty terms across all 73 persisted rules/checklists.

6.4 Three-Paradigm Blind Clinician Judge Concordance ($n = 191$ Findings)

Table 6 evaluates whether self-evolution depends on same-model bias by comparing three blind clinician paradigms on the exact same 191 Round 1 findings (114 ground-truth errors and 77 false positives).

Across all three paradigms, unanimous 3/3 agreement reaches **84.3**% (161/191, mean pairwise agreement 89.5%), yielding substantial 3-rater Fleiss' $\kappa = \mathbf{0.642}$ ($z = 7.85, p < 10^{-14}$; $\kappa = 0.657$ on `dev`, $\kappa = 0.618$ on `test`, and $\kappa = 0.679$ on ground-truth FPs). Notably, `deepseek-chat` exhibits near-perfect concordance with `deepseek-v4-flash` ($\kappa = 0.932$), and under cross-model `deepseek-chat` replies, intra-manuscript Round 2 cuts FPs by **−35.1**% (77 → 50, exact 2^{19} $p = 0.0073$, lifting Prec_{inj} +9.6 pp, $p = 0.0051$) with zero significant recall change ($p = 1.00$). Meanwhile, explicit CoT verification in `deepseek-reasoner` achieves the highest true-error acceptance (**97.4**%, rescuing 6 subtle arithmetic errors falsely rebutted by non-CoT judges, which accounts for moderate pairwise $\kappa = 0.500$ [85.9% raw agreement, PABAK = 0.717] under 83% positive prevalence), with **zero (**0/114**)** true errors unanimously falsely rebutted.

6.5 Red-Team Adversarial Audit, Pass A `<table-wrap>` Fidelity, and Track B Guards

Table 7 and Figure 4 present our three dedicated safety and component sub-benchmarks.

Table 7: Dedicated Red-Team & Component Sub-Benchmarks: **(A)** Full Hybrid-LLM Adversarial Rebuttal Stress Test ($N = 19$ Papers, 57 Errors, 120 Rebutted Injected-Paper Findings); **(B)** Pass A External Literature `<table-wrap>` Citation Benchmark ($N = 36$ Paired Claims across 12 PMC XMLs); and **(C)** Track B Pre-Write Procedural Revision Guard ($N = 24$ Unseen Tasks).

Sub-Benchmark & Regime	Configuration / Guard	Primary Accuracy / Recall [95% CI]	Key Safety / Error Metric	Secondary Precision / Subset	Exact Paired Test
Panel A: Full Hybrid-LLM Adversarial Rebuttal Stress Test ($N = 19$ **Papers, 57 Injected Errors, `acceptedFix=false` on Every Finding**)					
Round 1 Hybrid Cold-Start	`heurion_deep (useLlm=true)`	Recall: 84.2% (48/57) [72.6, 91.5]	`test` Recall: 83.3% (20/24)	Clean/Tot FP: 52/86 ($F1_{org} = 68.0\%$)	Reference R1
Pre-Hardening R2 Intra-Doc	Subtype-Only $\mathcal{F}_{\text{hard}}$	Recall: 71.9% (41/57) [59.2, 81.9]	23 **False `rev_exempt` (10 TPs Lost)**	`test` Collapse: **62.5**% (−20.8 pp)	Exposed `subtype=null` gap
Post-Hardening R2 Intra-Doc	**5-Invariant Hybrid $\mathcal{F}_{\text{hard}}$**	**Recall: 82.5**% (47/57) [70.6, 90.2]	**0 False `rev_exempt` (64/64 Blocked)**	Clean FP −**61.5**% ($F1_{org} =$ **74.1**%)	`dev` 84.8%, `test` 79.2%
Post-Hardening R2 Cross-Paper	Frozen Adversarial Track A	Recall: 82.5% (47/57) [70.6, 90.2]	**0/20** Leaked Digits/Terms	`dev` Recall: 87.9% (29/33)	97.9% R1 retention
Panel B: Pass A External Literature Citation Verification ($N = 36$ **Claims:** 12 **Abstract,** 12 **PMC `<table-wrap>`-Only,** 12 **Distorted**)					
`abstract_only_pass_a`	PubMed Title + Abstract	Verdict Acc: 52.8% (19/36)	Table-Only Support: **0.0**% (0/12)	Distortion Recall: 66.7% (8/12)	100% table false-unclear
`full_pass_a_with_pmc_tables`	**Abstract + JATS `<table-wrap>`**	**Verdict Acc: 94.4**% (34/36)	**Table-Only Support: 91.7**% (11/12)	**Distortion Recall: 100**% (12/12)	**McNemar** $p = 2.75 \times 10^{-4}$
Panel C: Track B Pre-Write / Revision Procedural Guard ($N = 24$ **Unseen Clinical Co-Authoring Tasks across 4 Families**)					
Cold-Start Authoring	`procedural = off`	Clean-Pass (Hyb): 58.3% (14/24) [38.8, 75.5]	Total Hybrid Errors: 15 (0.63/doc)	Clean-Pass (AST): 70.8% (17/24)	Flow/Math Clean: 0/6 (0%)
Track B Guarded Authoring	`procedural = utility` (3 **Rules**)	**Clean-Pass (Hyb): 87.5**% (21/24) [69.0, 95.7]	**Total Hybrid Errors: 4 (−73.3**%)	**Clean-Pass (AST): 91.7**% (22/24)	**McNemar** $p = 0.0156$ **(0 reg.)**

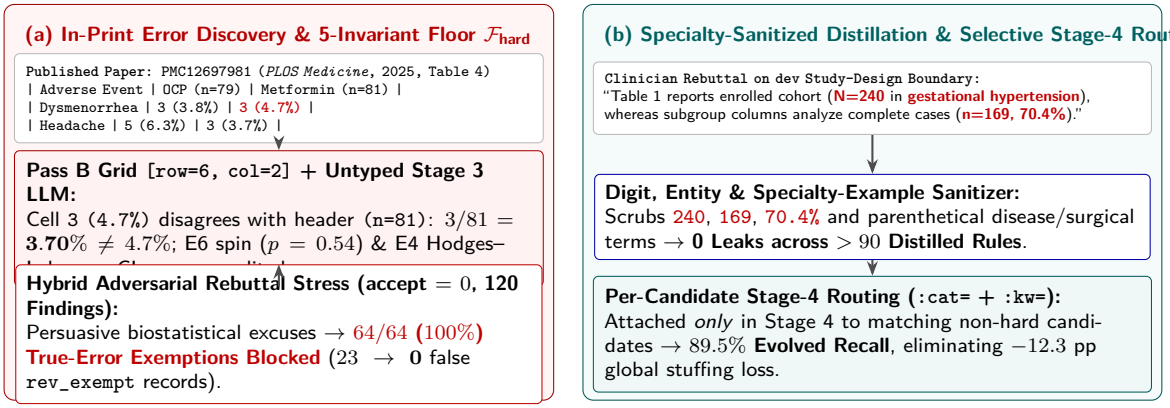


Figure 4: Qualitative safety and governance mechanisms. **(a)** Pass B catches a published Table 4 typo in *PLOS Medicine* (`PMC12697981`), and the hardened 5-invariant hybrid safety floor $\mathcal{F}_{\text{hard}}$ blocks 100% (64/64) of adversarial intra-document exemptions across both AST and untyped Stage 3 LLM findings. **(b)** Track A strips both raw digits and parenthetical disease/surgical examples before routing rules selectively per Stage 4 candidate.

1. Closing the Untyped Hybrid-LLM Adversarial Loophole (Panel A). When an adversarial author forcefully rebuts all 120 hybrid findings with plausible biostatistical rationales (e.g., claiming Hodges–Lehmann non-parametric CIs need not contain the median difference or $p = 0.54$ reflects exploratory trend framing), subtype-only gating protected 100% of deterministic AST alerts but allowed 23 untyped `llm_stage3` true positives to write `rev_exempt` records, dropping intra-document Round 2 recall from 84.2% to 71.9% (83.3% → 62.5% on `test`). Hardening $\mathcal{F}_{\text{hard}}$ with our five domain-general invariants eliminates 100% **of false intra-document exemption records (23 → 0; 64/64 true-positive findings guarded)**, restoring Round 2 recall to **82.5**% (84.8% on `dev`, 79.2% on `test`; ±1 item from section-level LLM context variance) while still suppressing **61.5**% of clean-paper false alarms (52 → 20, $F1_{org} = 74.1\%$).

2. Pass A PMC `<table-wrap>` Citation Fidelity ($N = 36$, Panel B) & Track B Pre-Write Guard ($N = 24$, Panel C). In external citation checking across 12 real 2025 PMC references ($N = 36$), abstract-only verification resolves **0.0**% (0/12) of faithful claims grounded in reference tables (flagging 12/12 as `unclear`) and catches only 2/6 (33.3%) of table-locus distortions (52.8% overall verdict accuracy). Parsing JATS XML `<table-wrap>` grids lifts table-only faithful support to **91.7**% (11/12, McNemar $p = 9.77 \times 10^{-4}$), distortion recall to **100.0**% (12/12), and overall verdict accuracy to **94.4**% (34/36, McNemar $p = 2.75 \times 10^{-4}$). Complementarily, in downstream clinical revision across 24 unseen tasks (0 test signals used; 3 zero-digit Track B rules crystallized from 38 `dev` fix signals), `ProceduralHarnessService` lifts the live hybrid zero-defect pass rate from 58.3% (14/24) to **87.5**% (21/24, +29.2 pp, exact McNemar $b = 7, c = 0, \mathbf{p = 0.0156}$, zero regressions) and cuts total hybrid errors by **−73.3**%

(15 → 4; sample-flow/arithmetic pass rate 0/6 → 5/6). Secondary cross-domain evaluation on the 7 clinical oncology vignettes (`deepseek-chat`, 8 organic errors) further confirms these trends: `heurion_deep` catches 100% (8/8) vs. 37.5% (3/8) for `zero_shot` and `pat_baseline`, and `heurion_evolved` eliminates all 16 recurring TTE clone-pseudocohort and translational boundary false alarms (0.0% leakage).

7 Discussion, Limitations, and Conclusion

Our experiments across 15,682 live API calls establish three design principles for trustworthy scientific review agents: (i) **2D structural AST and quote-verified LLMs are strictly complementary**, achieving precision and token efficiency unobtainable by single-pass or 64k-CoT brute force; (ii) **agentic memory must be routed selectively at verification time rather than globally stuffed into discovery prompts**, preventing conservative sensitivity shift while preserving order-invariant rule convergence; and (iii) **adversarial safety floors in hybrid systems must enforce mathematical, CI-geometric, and semantic-spin invariants over untyped LLM rationales**, not merely AST tags.

Limitations and Future Work. First, while our pipeline audits full-text prose, multi-column 2D tables, captions, and PMC `<table-wrap>` XMLs, it does not yet extract optical coordinates from rasterized Kaplan–Meier curves, forest plots, or Western blots. Second, Pass A full-table verification requires open-access PMC XML or user-uploaded reference PDFs; closed-access citations without full text default to abstract-tier coverage warnings. Third, although 5-fold cross-validation (57 errors) and secondary vignettes confirm robust architectural gains, expanding the controlled full-text benchmark to multi-hundred-paper specialty registries and conducting prospective blinded randomized reader studies with practicing physician-investigators are essential next steps. Finally, cloud MoE inference exhibits minor (±1–2 item) batching variance on borderline semantic findings even at `temperature = 0`, motivating deterministic ensemble voting in Stage 4 for high-stakes journal screening.

Conclusion. Heurion Deep Review unifies clean-room 2D table-grid AST verification, quote-verified four-stage hybrid auditing, candidate-scoped dual-track self-evolution, and a five-invariant hybrid safety floor—eliminating recurring clinical study-design false alarms, catching published errors in $\sim$ 20 s/manuscript, and preventing defects during clinical revision without compromising objective error recall.

Ethics Statement

All Benchmark 1 and Pass A manuscripts were retrieved from open-access PubMed Central (PMC) under Creative Commons licenses, and vignette excerpts were derived from published oncology literature. No private patient-identifiable health information or EHR data were accessed; IRB approval was not required.

Data and Code Availability

The complete clean-room verification engine (`apps/platform/src/claims/`), 19-paper corpus and error injections (`apps/platform/eval/deep-review/`), and all four live benchmark suites and raw JSON telemetry (`paper-deep-review/benchmarks/` and `paper-deep-review/results/`) are publicly available at https://github.com/0xSynapse/heurion.

Competing Interests

H. Zhao and the Heurion Research Team develop the Heurion clinical research platform. D. Qian declares no competing financial interests.

References

[1] Kenneth F. Schulz, Douglas G. Altman, and David Moher. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *BMJ*, 340:c332, 2010.

[2] Jan P. Vandenbroucke, Erik von Elm, Douglas G. Altman, Peter C. Gøtzsche, Cynthia D. Mulrow, Stuart J. Pocock, Charles Poole, James J. Schlesselman, and Matthias Egger. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *PLoS Medicine*, 4(10):e297, 2007.

[3] Isabelle Boutron, Susan Dutton, Philippe Ravaud, and Douglas G. Altman. Reporting and interpretation of randomized controlled trials with statistically nonsignificant results for primary outcomes. *JAMA*, 303(20):2058–2064, 2010.

[4] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia C. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021.

[5] Patrick M. Bossuyt, Johannes B. Reitsma, David E. Bruns, Constantine A. Gatsonis, Paul P. Glasziou, Les Irwig, Jeroen G. Lijmer, David Moher, Drummond Rennie, Henrica C. W. de Vet, et al. STARD 2015: an updated list of essential items for reporting diagnostic accuracy studies. *BMJ*, 351:h5527, 2015.

[6] Joseph L. Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971.

[7] DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, et al. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.

[8] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

[9] Rajesh Jayaram, Drew Tyler, David Woodruff, Corinna Cortes, Yossi Matias, Vahab Mirrokni, and Vincent Cohen-Addad. Towards automating scientific review with Google's Paper Assistant Tool. *arXiv preprint arXiv:2606.28277*, 2026.

[10] Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Collin Hinks, Michael J. Hammerling, Manvitha Ponnapati, Samuel G. Rodriques, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024.

[11] Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[12] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[13] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[14] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[15] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[16] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research (TMLR)*, 2024.

[17] Mo Li, Zixin Yin, Ting Cao, and Yunxin Liu. Training skills like parameters via self-supervised semantic diffusion. *arXiv preprint arXiv:2607.27557*, 2026.

[18] Jingguang Li, Yebo Wu, Zuyi Guo, Kailang Ma, Xianjie Dai, Han Zheng, Benwang Chen, Li Li, Can Rong, and Heye Huang. CoEM: Empowering long-context reasoning with commit-on-evidence memory. *arXiv preprint arXiv:2609.36935*, 2026.

[19] Sean Hardesty Lewis, Zuyi Guo, Benwang Chen, Zirui Liu, Hongyi Lin, and Heye Huang. SafeVantage: Vantage-aware memory for reliable embodied decisions. *arXiv preprint arXiv:2609.36906*, 2026.

[20] Jie Ma, Zhipeng Qian, Yufei Ma, Zihan Liang, Jiayi Ji, Qingpeng Cai, Ben Chen, Peng Jiang, and Xiaoshuai Sun. Skill-V: Verifiable self-evolving skill library for interactive agents. *arXiv preprint arXiv:2610.11781*, 2026.

[21] Bo Tang, Yang Zhang, Guomian Zhuang, Wenqiang Wei, Gaoyang Zheng, Lindong Xie, Yanchao Tan, Feiyu Xiong, Qingyu Yang, Edward Chung, and Zhiyu Li. From memory to skills: Evidence-grounded co-evolution governance for long-horizon LLM agents. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[22] Haozhen Zhang, Quanyu Long, Jianzhu Bao, Tao Feng, Weizhi Zhang, Haodong Yue, and Wenya Wang. MemSkill: Learning and evolving memory skills for self-evolving agents. *arXiv preprint arXiv:2602.02474*, 2026.

[23] Haiming Tang, Xianjie Dai, Gujie Shao, Zuyi Guo, Jingguang Li, Kailang Ma, Yihong Tang, and Heye Huang. MemTransfer: Benchmarking memory beyond matched experience in embodied decision-making. *arXiv preprint arXiv:2609.32313*, 2026.